6 Probabilities and entropy

Having been led to the thermodynamic function entropy, $S(U, V, N)$, by macroscopic arguments, and discovered many of its properties, nevertheless its nature remains unclear.

We turn to a microscopic approach which will provide an intuitive picture for entropy. We will exploit macroscopic systems having very large number of atoms, typically Avogadro's number, $N_{\text{A}} \simeq 10^{23}$. This allows, as we shall see, the quantities measured in an experiment on a particular system to be very close to an average over many identical such systems.

Some terminology.
The macroscopic state of a system, specified by a small number of variables selected from $(U, S, T, p, V, \cdots)$, is a **macrostate**.

The particular arrangement of the atoms in, say, a gas, in terms of the position and momentum of *each atom*, defines the **microstate** of the system. Clearly many microstates correspond to a particular macrostate. Initially we will focus just on *position* variables, not the momenta.

Of course these statements are not *completely* tight: both the microstate and macrostate of the system will be time-dependent, the former to a large degree, the latter to a very small degree (as it is an average of very many contributions). We will focus on the limit of the number of particles $N \to \infty$, where the latter effect is negligible. Note some variables may be time-*in*dependent if the system is isolated, e.g. the energy, U, independently of the value of N.

The key assumption that the probabilities of the microstates corresponding to a particular total energy, U, are equally probable. This idea was due to Gibbs, and is still not proven in general to be true. There are some special cases where it is not true – e.g. the macroscopic example of the Moon going around the Earth has conserved angular momentum as well as energy, so we do not observe, for example, the Moon orbiting in a different plane.

6.1 Example 1: particles in a box

Consider N *distinguishable, noninteracting,* atoms inhabiting a rectangular box in two-dimensions (for simplicity of depiction), where we consider the total area divided into two equal left and right parts, denoted L and R, see Fig. (6.1) for a microstate for $N = 4$. There are N atoms

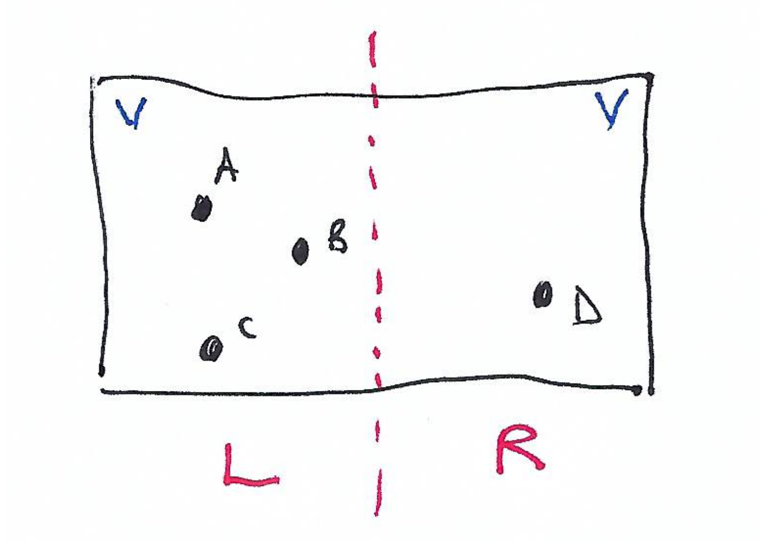

Figure 6.1: A box of volume $2V$ divided into two equal halves, each of volume V, denotes L and R. In this case there are four particles, labelled A-D.

and we choose the variable to define a macrostate to to be the number of atoms on the lefthand side, n. (The microstate, in terms of positions, would state the positions of each of the atoms.)

There are thus $N+1$ macrostates as the number on the lefthand side n obeys $0 \le n \le N$.

(i) **How many microstates are there?**
Each (distinguishable) atom can be on the L or R sides, so there are $\Omega = 2^N$ microstates in total.

(ii) **What is the probability of having all atoms on the lefthand side, p(all L)?**
Our assumption is that each microstate has the same probability, i.e.

$$p(\text{all } L) = \frac{1}{\Omega} = \frac{1}{2^N} \; .$$

(iii) **What is the probability of n atoms on the lefthand side?**
There are $\Omega(n) = {}^N C_n$ ways of choosing *which* n atoms are on the left (remembering they are distinguishable).

(iv) $\Omega(n)$ is termed the *statistical weight* of the macrostate labelled by n atoms on lefthand side. The probability of such a macrostate is

$$p(n) = \frac{\Omega(n)}{\Omega} = \frac{1}{2^N} \; {}^N C_n = \frac{1}{2^N} \frac{N!}{(N-n)!n!} \; . \tag{6.1}$$

We may derive the last result in another manner. Consider possible states an atom, A and B, to be exhaustive – i.e. an atom must be in one or the other. If the probability of an atom being in state A is p and the probability of being in state B is $q = (1-p)$, then the probability, $p(n; A)$, of n atoms being in state A is

$$\begin{aligned} p(n; A) &= {}^N C_n \, p^n q^{N-n} \\ &= {}^N C_n \, p^n (1-p)^{N-n} \; . \end{aligned}$$

If $p = q = \frac{1}{2}$, then find

$$p(n; A) = {}^N C_n \frac{!}{2^N} \; .$$

We will now examine how the expression Eq. (6.1) behaves as N becomes large. Fig. (6.2), Fig. (6.3) and Fig. (6.4) show the statistical weights, $\Omega(n)$ and the probability $p(n)$ for all possible macrostates, n, for $N = 4$, $N = 16$ and $N = 64$ atoms. These figures lead to two

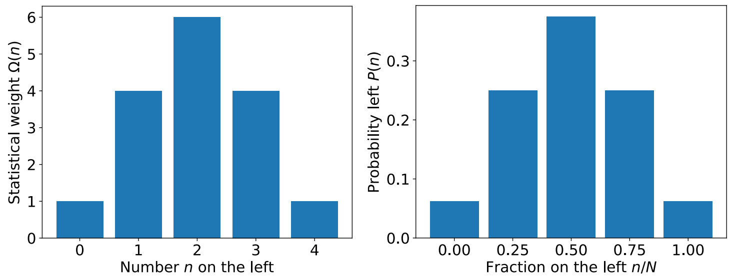

Figure 6.2: Left: statistical weights for macrostates with n atoms on the lefthand side, for $N = 4$. **Right:** probability for the same states as a function of the fraction of atoms on left, n/N.

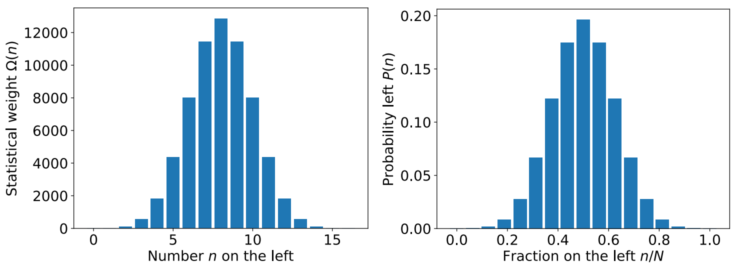

Figure 6.3: Left: statistical weights for macrostates with n atoms on the lefthand side, for $N = 16$. **Right:** probability for the same states as a function of the fraction of atoms on left, n/N.

observations:

(i) The distribution $\Omega(n)$ is centred at $n = N/2$, and the distribution $p(n)$ is centred at $n/N = \frac{1}{2}$. This corresponds to the mean value of n:

$$\overline{n} = \sum_{n=0}^{N} p(n) n = \frac{N}{2} \; .$$

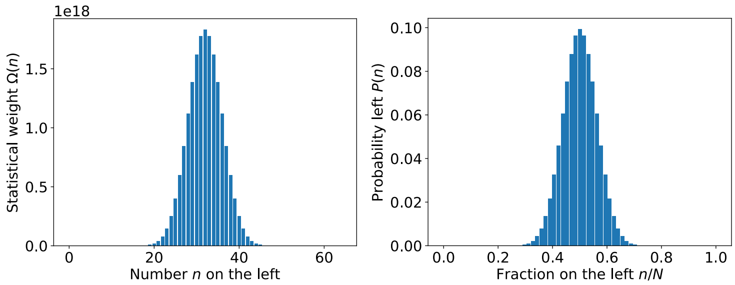

Figure 6.4: Left: statistical weights for macrostates with n atoms on the lefthand side, for $N = 64$. **Right:** probability for the same states as a function of the fraction of atoms on left, n/N.

(ii) The distributions become narrower as N becomes larger. The width of the distribution is characterised by the standard deviation, σ, for the binomial distribution:

$$\sigma = \left[\sum_{n=0}^{N} p(n)(n-\overline{n})^2\right]^{1/2} = \sqrt{Np(1-p)}$$
$$= \frac{\sqrt{N}}{2} \, .$$

If we examine the probability of the fraction of atoms on the left, $p(n/N)$, then it has a standard deviation σ_{f},

$$\sigma_{\mathsf{f}} = \frac{\sigma}{N} \quad \Rightarrow \quad \text{width distribution} \; = 2\frac{\sigma}{N} = \frac{1}{\sqrt{N}} \, .$$

To see the implication of these results in practice, let us examine some numbers:

No. of atoms	No. of macrostates	No. of microstates	$p(0)$ (all$R$)	$p(1/2)$ (50 / 50)
N	$N+1$	$\Omega = 2^N$	$\frac{1}{\Omega} = \frac{1}{2^N}$	${}^N C_n \, \frac{1}{2^N}$
4	5	16	0.0625	0.375
16	17	65536	1.5×10^{-5}	0.20
64	65	1.8×10^{19}	5.4×10^{-20}	0.10

It is clear that $p(0)$ decreases much more quickly with N than $p(1/2)$ does. In terms of timescales for the observation of all particles on L or R, consider the case of $N = 64$ in a box of dimension 1 metre with the atoms travelling at 100 ms^{-1}, a crude estimate for the time for the system to be in a new microstate is 10 ms (every atom has crossed the box). So given the number of microstates is $\simeq 2 \times 10^{19}$, it will take around 6×10^9 years to explore all microstates. This is around half the age of the Universe, so in a real experiment we will not observe all the atoms in one side or the other.

$N = 64$ might represent an excited ^{64}Ni nucleus. What about $N = N_{\rm A}$? Then the width of the peak around $n/N = \frac{1}{2}$ is $\sigma/(N/2) = 10^{-12}$, as in Fig. (6.5) The very important

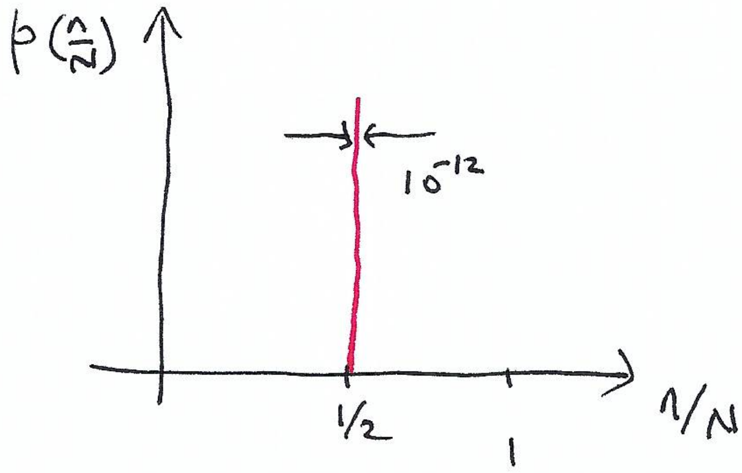

Figure 6.5: For $N = N_{\rm A}$, the width of the distribution $p(n/N)$ is extremely small!

deduction from this is that for typical macroscopic systems with $N_{\rm A}$ particles (even much smaller – like the number of particles for an excited ^{64}Ni nucleus), the overwhelmingly most probable macrostate is that which corresponds to the statistical average of the macroscopic variable (here n or n/N) over the accessible microstates. The fluctuations around that mean value are extremely small – here around 10^{-12}.

6.2 Entrance of entropy

Let us consider again the Joule experiment. Initially all the atoms are confined to L with volume V, by a real partition. In the framework of our crude model there is a single allowed microstate, $\Omega(V) = 1$ (corresponding to the macrostate $n = N$), whereas if we remove the partition the number of microstates is $\Omega(2V) = 2^N$.

When the partition is removed (assuming the gas is surrounded by an adiabatic container, so the temperature is unaltered), we know from classical thermodynamics (e.g. problem (2.4)), with n_{mol} the number of moles not the number on L)

$$
\begin{aligned}
\Delta S &= n_{\text{mol}} \mathsf{R} \ln\left(\frac{V_{\mathsf{f}}}{V_{\mathsf{i}}}\right) > 0 \\
&= N \mathsf{k}_{\rm B} \ln\left(\frac{2V}{V}\right) \\
&= \mathsf{k}_{\rm B} \ln\left(2^N\right) \qquad \text{which may be written} \\
\Delta S &= \mathsf{k}_{\rm B} \ln\left(\frac{\Omega(2V)}{\Omega(V)}\right) \\
&= \mathsf{k}_{\rm B} \ln \Omega(2V) - \mathsf{k}_{\rm B} \ln \Omega(V) \ .
\end{aligned}
$$

This suggests that $S = \mathsf{k}_{\mathrm{B}} \ln \Omega$ where Ω is the statistical weight of the macrostate. This is Boltzmann's equation for entropy. It provides an intuitive picture for entropy: a macrostate corresponding to more microstates, has more "disorder" (one does not know which microstate the system is in at any given time) and this corresponds to a larger entropy.

So if we consider the macrostates for n atoms of the gas being in L, then

$$\begin{aligned} S(n) &= \mathsf{k}_{\mathrm{B}} \ln \Omega(n) \\ \Rightarrow \quad S(n=0) = S(n=N) &= \mathsf{k}_{\mathrm{B}} \ln 1 = 0 \\ \text{and} \quad S\left(\frac{N}{2}\right) & \text{ is maximal.} \end{aligned}$$

Returning to Joule's experiment, the gas is released from a state with low entropy and ends with a much larger entropy, illustrating the increase of entropy expected in irreversible processes.

6.3 Example 2: the paramagnet

The following model has the simplification that we may describe the energy of the states of the system almost precisely. In the gas example at the beginning of the section, we did not consider the energy of the gas atoms (and hence could not discuss temperature, and the temperature dependent part of the entropy derived in problem (2.4)). The macrostates were also not equilibrium macrostates. These issues will be addressed more satisfactorily with this second example.

Consider an array of particles with spin-1/2 with an associated magnetic moment $\boldsymbol{\mu}$, which for the case of an electron is

$$\boldsymbol{\mu} = -g_{\mathrm{S}} \mu_{\mathrm{B}} \frac{\mathbf{s}}{\hbar} \quad \text{and} \quad \mu_{\mathrm{B}} = \frac{e\hbar}{2m_{\mathsf{e}}} \,, \quad \text{(SI units)} \tag{6.2}$$

where $g_{\mathrm{S}} \simeq 2$, μ_{B} is termed the *Bohr magneton* the characteristic scale of the magnetic moment, e is the (modulus of the) charge of the electron and m_{e} is the mass of the electron. The deviation from $g_{\mathrm{S}} = 2$ (predicted by the Dirac equation) comes from QED effects at the 0.1% level.

We assume the particles are embedded in a non-magnetic host material and are very dilute, so we do not need to describe effects of their dipole-dipole interaction (this is quite subtle, and we will ignore). Thus they only interact with an external applied magnetic field, $\mathbf{H}$.

Then the energy of interaction of each magnetic moment with the external field is

$$U \simeq -\mu_0 \boldsymbol{\mu} \cdot \mathbf{H} \simeq -\boldsymbol{\mu} \cdot \mathbf{B} \,, \tag{6.3}$$

where $\mathbf{B} = \mu_0 \mathbf{H}$ assumes the fields due to the other dipoles are negligible. Given there are only two states for a spin-1/2 particle, we find $U = \pm \mu B$, the minus sign when the magnetic moment is parallel to $\mathbf{B}$, and the positive sign when they are antiparallel. The separation in energy of the two states for each moment is $2\mu B$.

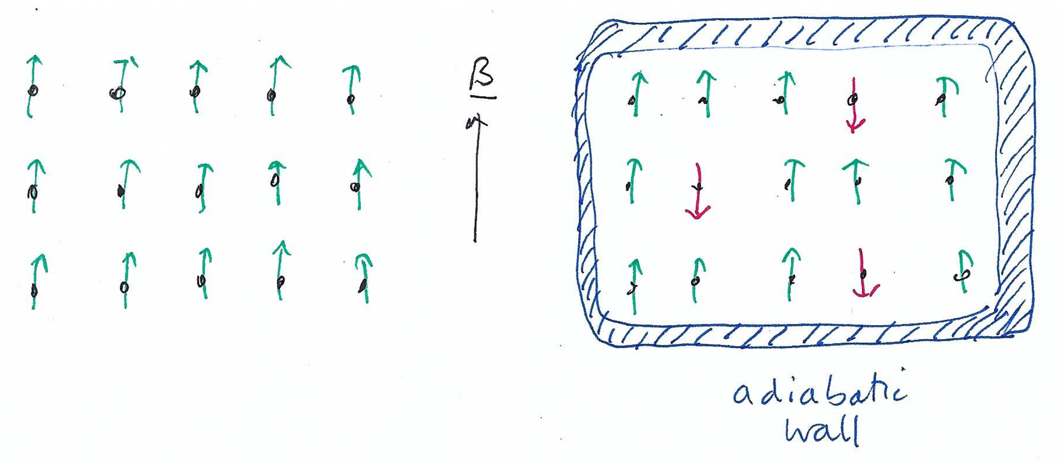

Figure 6.6: Left: The ground state, with all moments parallel to the field, ↑. **Right:** three moments, ↓, have been flipped, with the addition of $6\mu B$ of energy.

Consider a small 3×5 lattice, depicted in Fig. (6.6) in its ground state on the left with all moments parallel to the field, ↑, with energy $U_{\text{tot}} = -15\mu B$.

If we now add $6\mu B$ of energy, flipping three moments to ↓, so $U_{\text{tot}} = -9\mu B$. Fig. (6.6) shows one microstate corresponding to this energy. There are ${}^{15}C_3 = 455$ possible microstates with three flipped moments (and hence this energy), with an associated entropy

$$S = \mathrm{k_B} \ln \left({}^{15}C_3\right) = 8.4 \times 10^{23}\ \mathrm{JK}^{-1} ,$$

while the ground state has entropy $S_0 = \mathrm{k_B} \ln 1 = 0$.

For the general case of N moments with n flipped ↓ moments, the total magnetisation is

$$\begin{aligned} M^{\text{tot}} &= (N - 2n)\mu \\ \Rightarrow \quad U_{\text{tot}} &= -M^{\text{tot}} B \\ \text{and} \quad S(n) &= \mathrm{k_B} \ln \left({}^{N}C_n\right) . \end{aligned}$$

Using Stirling's approximation, $N! \simeq (N/\mathrm{e})^N$, we can see that $S(n = N/2) \simeq N \ln 2$, i.e. each moment contributes $\ln 2$ to the entropy.

Note. In Electromagnetism 2 next semester, the magnetisation is defined to be a vector, $\mathbf{M}$, which has the magnitude of M^{tot} above divided by the volume of the system, i.e.

$$\mathbf{M} = \frac{M^{\text{tot}}}{V} \hat{\mathbf{z}} .$$

The volume is the *total* volume, including the non-magnetic material as well.

6.4 Heat exchange and thermalisation

We will now discuss a microscopic and statistical description of the process of heat exchange and thermalisation.

Consider two initially independent systems A and B, which we wish to describe as a single entity. Let A be in a (macro)state with statistical weight Ω_A and B has Ω_B. Then, since the systems are independent, the statistical weight of the combined system, $A + B$, is

$$\begin{aligned} \Omega_{A+B} &= \Omega_A \times \Omega_B \\ \Rightarrow \quad S_{A+B} &= \mathsf{k}_\mathrm{B} \ln\left(\Omega_{A+B}\right) \\ &= \mathsf{k}_\mathrm{B} \ln\left(\Omega_A \times \Omega_B\right) \\ &= \mathsf{k}_\mathrm{B} \ln \Omega_A + \ln \Omega_B \\ \Rightarrow \quad S_{A+B} &= S_A + S_B \; . \end{aligned} \tag{6.4}$$

So entropy adds and hence is extensive.

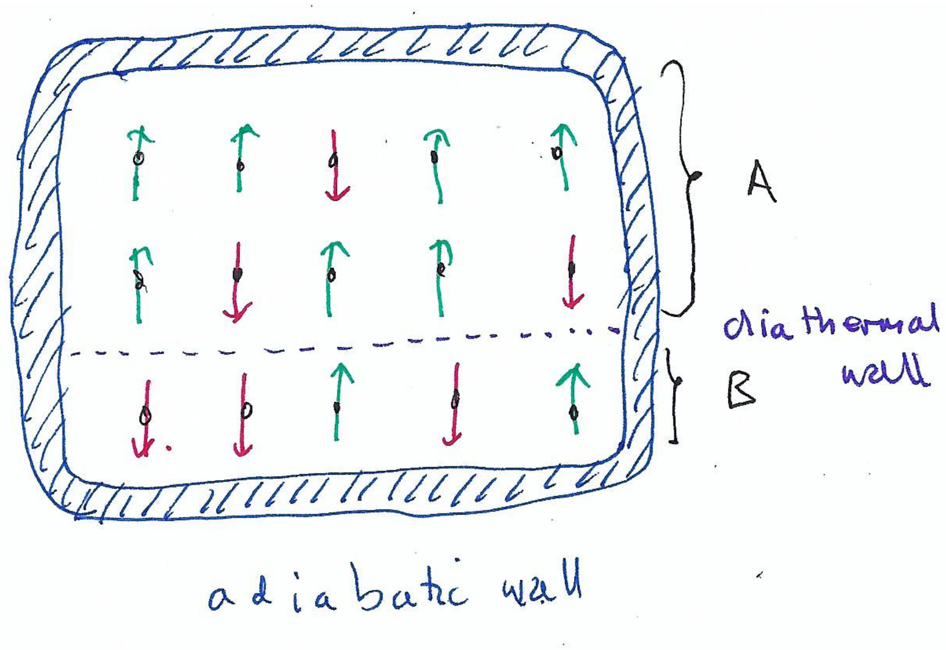

Figure 6.7: 10 sites of system A and five sites of system B, with a diathermal wall between them, allowing exchange of energy.

Now consider a system A with 10 magnetic moments and B with 5, see Fig. (6.7). Initially there are 6 moments $\downarrow$, three each in A and B. Thus the combined system has an energy $12\mu B$ above the ground state. The statistical weight of the initial state is

$$\Omega_{A+B} = \Omega_A \times \Omega_B = {}^{10}C_3 \times {}^{5}C_3 = 120 \times 10 = 1200 \; .$$

Let us now flip one extra moment in A and convert one flipped moment in B back to being parallel to the field; there are now 4 flipped moments in A and 2 in B. This process does not alter the total energy. The new statistical weight is

$$\Omega'_{A+B} = \Omega'_A \times \Omega'_B = {}^{10}C_4 \times {}^5C_2 = 210 \times 10 = 2100 \ ,$$

meaning this state is more probable, as long as the energy is allowed to flow between the systems A and B, as clearly the energy of the two systems does change although the total energy is fixed. Table (6.1) and Fig. (6.8) show the statistical weight for all microstates with

n	m	${}^{10}C_n$	${}^5C_m$	${}^{10}C_n \times {}^5C_m$
6	0	210	1	210
5	1	252	5	1260
4	2	210	10	2100
3	3	120	10	1200
2	4	45	5	225
1	5	10	1	10

Table 6.1: Statistical weights of all macrostates with exactly 6 moments down.

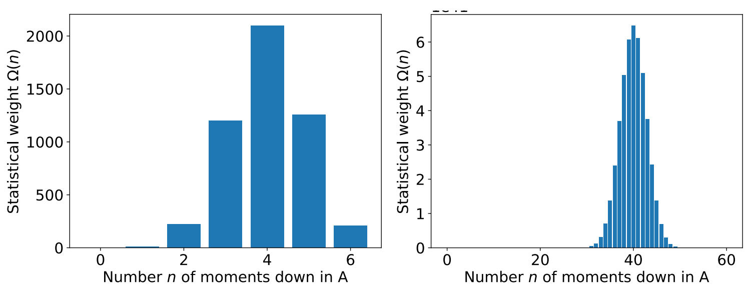

Figure 6.8: Statistical weights of all macrostates with exactly 40% of moments down. **Left:** 15 moments in total. **Right:** 150 moments in total, vertical scale in units of 10^{41}.

energy $12\mu B$ above the ground state energy. One can see that the average number of moments which are down in A is close to 4 (which also corresponds to the most probable macrostate). This means that when energy is allowed to flow between A and B, the distribution of down moments between A and B is close to $\frac{2}{3}/\frac{1}{3}$ on average, after starting with $1/1$.

In this case it is not surprising as A and B are made of the same microscopic constituents implying equilibrium corresponds to equal energy densities.

The energy which has been exchanged between A and B (in this case from B to A) is heat. This microscopic model provides a description of what heat is: a transfer of energy at a microscopic level, where particles change state.

What happens in the limit of a large number of particles (the *thermodynamic limit*)? The righthand figure in Fig. (6.8) shows the same distribution of statistical weights when there are 100 moments in A and 50 moments in B, for an identical energy density, i.e. a total of 60 moments are down. One can see the average number of down moments in A remains the same (when normalised by the number of moments), but the distribution becomes relatively narrower. At typical numbers, N_{A}, it appears that macrostates which have nearly equal energy densities in A and B are overwhelmingly more probable and the equalisation of energy densities becomes a fact.

Recollect Example 1, where we considered particles moving between left- and right-hand sides of the box. In the limit of large numbers of particles there was a vanishing probability of all particles being in one side of the box. Here the flipped moments play the role of the atoms, with A and B corresponding to L and R (albeit with volumes $2/3V$ and $1/3V$. Again as the size of the system becomes large (with the number of flipped moments growing at same rate), it becomes increasingly unlikely to see all the flipped in moments in A.

In Example 1 we did not concern ourselves with the dynamics of the particles except when we estimated how long one would need to wait for all the particles to be in one half.

One might ask, nevertheless, what the counterpart for the paramagnet system which allows the spin flips to move. If we add to the energy the second term below with spin raising and lowering indices $s^{\pm} = s^x \pm \mathrm{i}s^y$,

$$\mathcal{E} = \frac{g_{\mathrm{s}}\mu_{\mathrm{B}}}{\hbar} B \sum_{i=1}^{N} s_i^z - J \sum_{i=1}^{N} \sum_{j=1}^{N} \left(s_i^+ s_j^- + s_i^- s_j^+ \right) \ , \tag{6.5}$$

with $N^2 J \ll 2\mu B$, then the second term allows the flips to move but does not significantly alter the energy of the system, due to the factor of N^2 in the inequality.

6.5 Temperature

Let $S(U, U_A)$ be the entropy of the macrostate that has exactly energy U_A in A and total energy U. Therefore the energies of both A and B are defined from Eq. (6.4)

$$\begin{aligned} S(U, U_A) &= \mathsf{k}_{\mathrm{B}} \ln \left(\Omega_A \Omega_B \right) \\ &= S_A(U_A) + S_B(U_B) \ . \end{aligned}$$

$S(U, U_A)$ is maximised for the most probable state, i.e. the state with the highest statistical weight in Fig. (6.8). Mathematically

$$\begin{aligned} \frac{\partial S(U, U_A)}{\partial U_A} = 0 &= \frac{\partial S_A(U_A)}{\partial U_A} + \frac{\partial S_B(U_B)}{\partial U_A} \\ &= \frac{\partial S_A(U_A)}{\partial U_A} - \frac{\partial S_B(U_B)}{\partial U_B} \qquad \text{(Using } U_B = U - U_A\text{.)} \\ \Rightarrow \quad \frac{\partial S_A(U_A)}{\partial U_A} &= \frac{\partial S_B(U_B)}{\partial U_B} \ . \end{aligned} \tag{6.6}$$

Remembering from the First law,

$$\frac{1}{T} = \left(\frac{\partial S}{\partial U}\right)_{V,N} ,$$

we may identify in Eq. (6.6)

$$\frac{\partial S_A(U_A)}{\partial U_A} = \frac{1}{T_A} = \frac{1}{T_B} = \frac{\partial S_B(U_B)}{\partial U_B} ,$$

implying $T_A = T_B$ for the state in the middle of the peak.

6.6 Temperature of a paramagnet as a function of energy

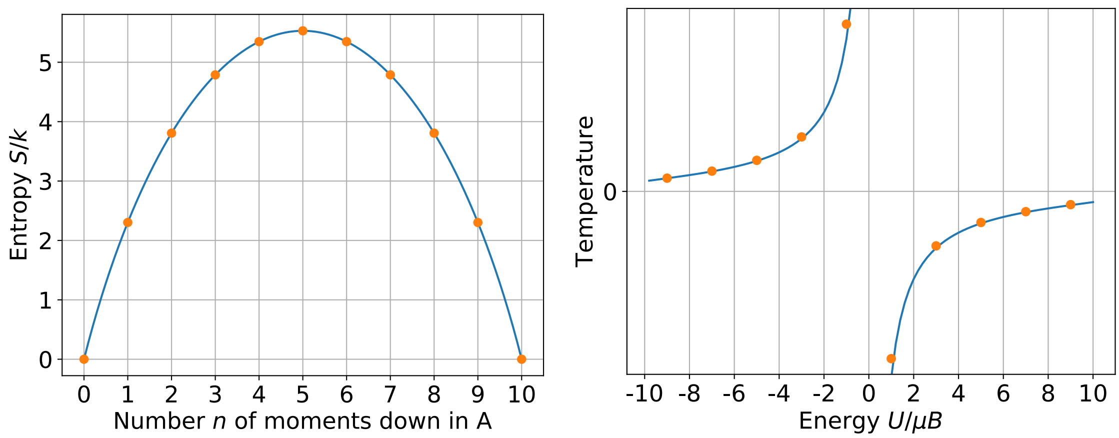

Figure 6.9: Left: Entropy of a paramagnet of 10 moments as a function of the number of flipped moments. **Right:** Temperature of the paramagnet as a function of the internal (magnetic) energy. Note that the x-scales for the two plots are equivalent. The blue curves are continuous extrapolation.

Remember the notation x^+ means approaching x from values above x and x^- means approaching x from values below x.

Consider the left figure in Fig. (6.9) in the context of $T^{-1} = (\partial S/\partial U)_N$. We can see that as $U \to 0^-$, $T^{-1} \to 0^+$ i.e. $T \to +\infty$. and for $U \to 0^+$ (i.e. from the right in the figure) $T^{-1} \to 0^-$ or $T \to -\infty$! So $T = \pm\infty$ are actually very close to each other. For all $U > 0$, $T < 0$, and $T \to 0^-$ as $U \to U^{\text{max}} = 10\mu B$. thus while $T = \pm\infty$ are very similar – equal probabilities of moments being down or up – $T = 0^\pm$ are very different. For $T = 0^+$, all moments are parallel to the field, in the lowest energy state, $U^{\text{min}} = -10\mu B$, while for $T = 0^+$, the moments are *anti-parallel* to the field with $U^{\text{max}} = 10\mu B$.

If $U > 0$, i.e. $T < 0$, then the system can increase its entropy by donating energy to the environment. Thus to maximise entropy it will lose energy until it is at $T = \infty$, and then to equilibrate to the (presumably) positive temperature of its surroundings, losing entropy.

Note the possibility of negative temperatures only occurs when there is a maximum energy for a system (in this case all moments anti-parallel to the magnetic field). So a gas with kinetic energy cannot experience this, as there is no upper limit to the kinetic energy.

Negative temperatures can be created in systems with maximum energies and also very weakly coupled to their surroundings. This was originally achieved with *nuclear* moments in a crystal which interact with each other by dipole-dipole interactions, but are weakly coupled to the phonons in the crystal.

6.7 Isothermal systems

We now consider the arrangement of the last subsection, with two systems weakly coupled thermally, to the asymmetric limit of one having a much larger thermal capacity than the other. We will term the small system the "system", A, and the large system, B, a thermal "bath", as it can absorb or gain energy from A without its temperature varying. The although the temperature of A is fixed, to that of the bath, B, with which it is in equilibrium, the energy of A, U_A, may vary.

What is the probability of A being observed to have energy E, i.e. $U_A = E$? To answer this for the case where the energy levels are those of a small system the notion of **degeneracy** is important, whether a given energy **level** corresponds to more than one **state**. We define the degeneracy by saying level E_n, with $n = 0, 1, \cdots$, corresponds to $g_n \geq 1$ states. If $g_n = 1$, we say that level n is **non-degenerate**. Comparing with earlier definitions, g_n corresponds to the statistical weight of the system having $U = E_n$.

Example: discrete, consider two atoms, labelled A and B, which both may reside in either of two energy levels 0 and $\varepsilon > 0$, see Fig. (6.10). .The notation in the middle of the figure is (energy of atom A, energy of atom B).

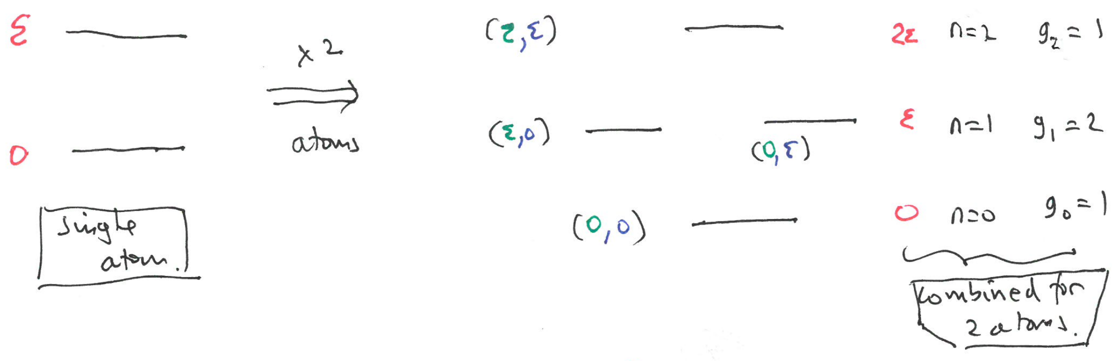

Figure 6.10: Left: energy levels for **one** atom. **Right:** energy levels, E_n (with $n = 0$, 1 and 2), for the combined system of two atoms, green and blue, and their degeneracies, g_n.

Continuous case: in classical systems or when the spacing between levels is very small a continuous description is needed or desirable (respectively). Then we take a *density of states*, $g(E)$, where $g(E)\mathrm{d}E$ is the number of states with energy lying between E and $E + \mathrm{d}E$.

6.7.1 Boltzmann distribution

Returning to the original problem of a "small" system, A, in thermal contact with a bath, B, with large thermal capacity. The composite of A and B is thermally isolated from their surroundings. the system is depicted in Fig. (6.11).

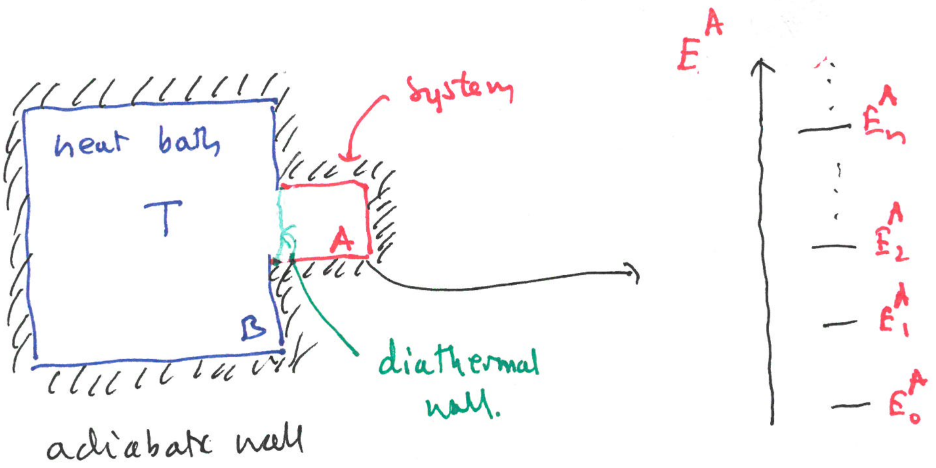

Figure 6.11: **Left:** heat bath, B, and system, A, isolated from their environment, but with a diathermal wall between them allowing energy exchange. **Right:** energy levels, E_n^A, for the system, A.

What is the probability of the system being in a state energy with E_i?

We will apply the Boltzmann expression for entropy to the combined system. The total energy U_0 is fixed, and let us assume that A is in an energy **state** E_i^A. These two conditions define a macrostate with the statistical weight of B (as the state of A has been fixed):

$$\begin{aligned}\Omega(U_0, E_i) &= \Omega_B(U_0 - E_i^A) \\ &= \exp\left[\frac{S_B(U_0 - E_i^A)}{\mathsf{k}_B}\right]\end{aligned}$$

using

$$S_B(U_0 - E_i^A) = \mathsf{k}_B \ln \Omega_B(U_0 - E_i^A) \ .$$

We have assumed that the bath, B, is unaffected by A except in terms of conservation of energy.

Now, since $E_i^A \ll U_0$, as the bath has very large thermal capacity, we may Taylor-expand:

$$S_B(U_0 - E_i^A) \simeq S_B(U_0) - E_i^A \underbrace{\left(\frac{\partial S_B}{\partial U_B}\right)_{V,N}}_{=1/T} \tag{6.7}$$

$$= S_B(U_0) - \frac{E_i^A}{T} \tag{6.8}$$

$$\Rightarrow \quad \Omega = \exp\left[\frac{S_B(U_0)}{k_B} - \frac{E_i^A}{k_B T}\right] \tag{6.9}$$

$$\propto e^{-E_i^A/k_B T} . \tag{6.10}$$

Now the probability, p_i, of observation of the state with $U_A = E_i^A$, is proportional to the statistical weight, so

$$p_n \propto \Omega \quad \Rightarrow \quad p_i = \frac{e^{-E_i^A/k_B T}}{Z}$$

where Z is the normalisation (termed the *partition function*)

$$Z = \underbrace{\sum_{i=0} e^{-E_i^A/k_B T}}_{\text{sum over } \textbf{states}} = \underbrace{\sum_{n=0} g_n e^{-E_n^A/k_B T}}_{\text{sum over } \textbf{levels}} . \tag{6.11}$$

Since there are g_n states with energy $E_n^A = E_i^A$ (g_n may be 1), the probability of observing the energy **level** ^{A} is

$$p(E_n^A) = g_n p_i = \frac{g_n e^{-E_n^A/k_B T}}{Z} .$$

Here the g_n states within level n are mutually exclusive possibilities, so we may add their probabilities to get that of the level as a whole.

This expression is one of the most useful in the module!

Example 1
Consider an atom with 2 **non-degenerate** levels, $E_0 = 0$ and $E_1 = \varepsilon$. Then the partition function is (introducing $\beta = 1/k_B T$)

$$Z = \sum_{n=0}^{1} e^{-\beta E_n}$$

$$= e^{-\beta E_0} + e^{-\beta E_1}$$

$$= 1 + e^{-\beta\varepsilon} \tag{6.12}$$

$$\Rightarrow \quad p(E_0) = \frac{e^{-\beta E_0}}{Z} = \frac{1}{1 + e^{-\beta\varepsilon}} \tag{6.13}$$

$$\text{and} \quad p(E_1) = \frac{e^{-\beta E_1}}{Z} = \frac{e^{-\beta\varepsilon}}{1 + e^{-\beta\varepsilon}} \tag{6.14}$$

$$. \tag{6.15}$$

See lefthand bottom figure in Fig. (6.12).

We may use the Boltzmann distribution to produce averages of quantities, and by the previous development, for systems with many particles/moments those averages will have small variances and be the equilibrium, thermodynamic, expressions.

Consider a function of energy, $f(E)$, which has an average

$$\begin{aligned}\overline{f} &= \sum_n f(E_n)p(E_n) \\ &= \sum_n f(E_n)\frac{g_n}{Z}\mathrm{e}^{-E_n/\mathrm{k_B}T} \ .\end{aligned}$$

In the example above, Eq. (6.12), Eq. (6.13) and Eq. (6.14), we see if $f(E) = E$,

$$\begin{aligned}\overline{E} &= \frac{0 \times \mathrm{e}^{-0} + \varepsilon\mathrm{e}^{-\beta\varepsilon}}{Z} \\ &= \frac{\varepsilon\mathrm{e}^{-\beta\varepsilon}}{1+\mathrm{e}^{-\beta\varepsilon}} \ .\end{aligned}$$

There are some easy and revealing limits:

(i) $T \to 0^+$: $p(E_0) = 1$ and $p(E_1) = 0$, so the system is in its ground state.

(ii) $T \to \infty$: $p(E_0) = 1/2$ and $p(E_1) = 1/2$, so the states are equally probable.

(iii) $T \to 0^-$: $p(E_0) = 0$ and $p(E_1) = 1$, so the system is in its excited state.

Example 2

Let

$$E_0 = 0 \ , \ g_0 = 1 \quad \text{and} \quad E_1 = \varepsilon \ , \ g_1 = 3 \ ,$$

so the states are 0 -3, with

$$E_i = \begin{cases} 0 & \text{if } i = 0 \\ \varepsilon & \text{if } i = 1\ ,2\ ,3\ . \end{cases}$$

see top figure of Fig. (6.12).

So then the partition function may be written in terms of levels, $n = 0, 1$ or states, $i = 0, 1, 2, 3$:

$$
\begin{aligned}
Z &= \underbrace{\sum_{n=0}^{1} g_n \mathrm{e}^{-\beta E_n}}_{\text{levels}} \\
&= 1 \times 1 + 3 \times \mathrm{e}^{-\beta\varepsilon} \\
Z &= \underbrace{\sum_{i=0}^{3} \mathrm{e}^{-\beta E_i}}_{\text{states}} \\
&= 1 + \mathrm{e}^{-\beta\varepsilon} + \mathrm{e}^{-\beta\varepsilon} + \mathrm{e}^{-\beta\varepsilon} \\
&= 1 \times 1 + 3 \times \mathrm{e}^{-\beta\varepsilon}\ .
\end{aligned}
$$

The probabilities are shown in the bottom-right part of Fig. (6.12). Note that as $T \to 0^+$ the system is in its ground states, like the first example. But for $T \to \infty$, $p(E_1) = 3p(E_0)$, but the **states** are equiprobable, $p(\text{any } \textbf{state}) = 1/4$. In Y3 Statistical Physics you will see that

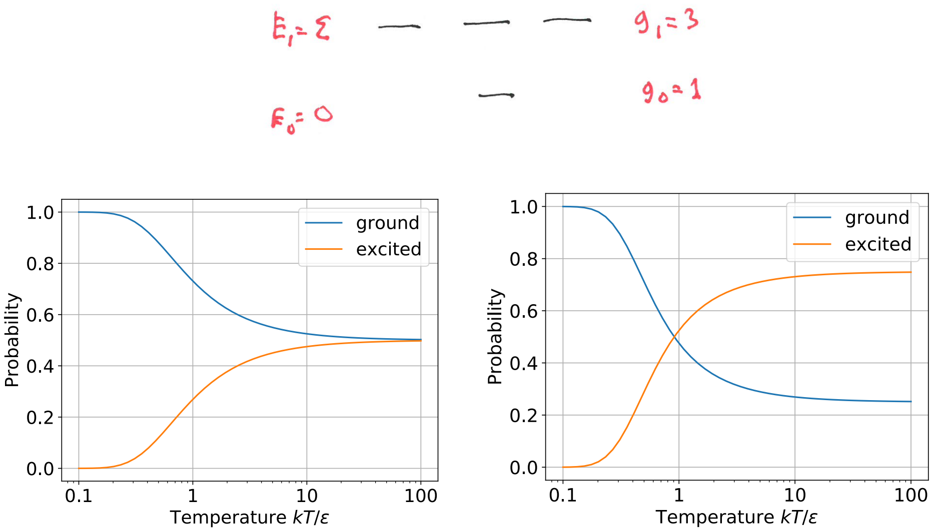

Figure 6.12: Top: the degeneracy, g_0 and g_1, of levels, $E_0 = 0$ and $E_1 = \varepsilon$. **Bottom left:** probabilities of occupations (populations) for a 2-level system with equal degeneracy as a function of the temperature. **Bottom right:** probabilities of occupations (populations) for a 2-level system with unequal degeneracy as a function of the temperature.

Z has much greater significance than merely the normalisation constant.

6.7.2 The continuous case

If we consider a classical system the energy is normally a continuous variable.[1] For a quantum system in a big box, $L \to \infty$, the energy levels are proportional to L^{-2}, so become very close together and one may make a "continuum limit", replacing a set of closely spaced levels by a density of states, $g(E)$. Then the probability of the energy to be between E and $E + \mathrm{d}E$ is

$$p(E)\mathrm{d}E = \frac{g(E)}{Z}\mathrm{e}^{-\beta E} \tag{6.16}$$

$$\text{with} \quad Z = \int \mathrm{d}E \; g(E)\mathrm{e}^{-\beta E} \tag{6.17}$$

$$\text{where} \quad \overline{f} = \frac{1}{Z}\int \mathrm{d}E \; f(E)g(E)\mathrm{e}^{-\beta E} \; , \tag{6.18}$$

is the average of $f(E)$. The range of integration is over the whole range of allowed energies; normally there is a lower limit, but the upper limit may be infinity.

Example 1: a particle in a potential
Sometimes it is natural to parameterise the states not by the energy but by position, x, or momentum, p (for example in one-dimension). If the particle experiences a potential $V(x)$, and resides in the interval $a \le x \le b$, then the probability of lying between x and $x + \mathrm{d}x$ is

$$p(x)\mathrm{d}x = \frac{\mathrm{e}^{-V(x)/k_B T}}{Z}\mathrm{d}x$$

$$\text{and} \quad Z = \int_a^b \mathrm{d}x \; \mathrm{e}^{-V(x)/k_B T} \; .$$

A slight subtlety is that we assume the "container" is "flat" for $g(x) = 1$. If the particle was experiencing, for example, a gravitational potential and the particle resides on a surface whose elevation varies, $z(x)$, then (in one-dimension), the potential energy is $V(x) = mgz(x)$, where m is the mass of the particle and g is the gravitational acceleration. Then

$$g(x) = \sqrt{1 + \left(\frac{\mathrm{d}z}{\mathrm{d}x}\right)^2} \quad \Rightarrow \quad Z = \int \mathrm{d}x \; \sqrt{1 + \left(\frac{\mathrm{d}z}{\mathrm{d}x}\right)^2}\,\mathrm{e}^{-mgz(x)} \; ,$$

reflecting the arc-length of the section of surface corresponding to $\mathrm{d}x$ along the x-axis is larger than $\mathrm{d}x$. We are assuming "friction" ensures the particle stays at x when it is placed on the slope, $\mathrm{d}z/\mathrm{d}x \neq 0$!

Example 2: the Maxwell-Boltzmann distribution

We consider one-dimension. Here all the energy is kinetic, so $E = \frac{1}{2}mv^2$, so the probability

[1] But not always – e.g. balls residing on a stair case experiencing a gravitational potential. There the allowed energies are $E_n = nmgh$, where h is the ehight of each step.

of observing the particle between v and $v + \mathrm{d}v$ is

$$\begin{aligned} p(v)\mathrm{d}v &= \frac{1}{Z}\mathrm{e}^{-\beta m v^2/2}\mathrm{d}v \\ \text{where} \quad Z &= \int_{-\infty}^{\infty} \mathrm{d}v \ \mathrm{e}^{-\beta m v^2/2} \\ &= \sqrt{\frac{2\pi \mathsf{k}_\mathrm{B} T}{m}} \ , \end{aligned}$$

where we have used the standard Gaussian integral

$$\int_{-\infty}^{\infty} \mathrm{d}x \ \mathrm{e}^{-\alpha x^2} = \sqrt{\frac{\pi}{\alpha}} \ .$$

So the celebrated Maxwell-Boltzmann distribution is in one-dimension

$$p(v) = \sqrt{\frac{m}{2\pi \mathsf{k}_\mathrm{B} T}}\mathrm{e}^{-\beta m v^2/2} \ .$$

We may turn this into $p(E)$, using

$$v = \pm\sqrt{\frac{2E}{m}} \quad \Rightarrow \quad \frac{\mathrm{d}v}{\mathrm{d}E} = \frac{1}{2}\sqrt{\frac{2}{mE}}$$

so:

$$\begin{aligned} p(v)\mathrm{d}v &= p(v(E))\frac{\mathrm{d}v}{\mathrm{d}E}\mathrm{d}E \\ &= \sqrt{\frac{m}{2\pi \mathsf{k}_\mathrm{B} T}} \ \mathrm{e}^{-\beta E}\frac{1}{2}\sqrt{\frac{2}{mE}} \ \mathrm{d}E \\ &= \frac{1}{2}\frac{1}{\sqrt{\pi \mathsf{k}_\mathrm{B} T E}} \ \mathrm{e}^{-\beta E}\mathrm{d}E \ . \end{aligned}$$

Notice this expression is dimensionless – as it should be, to integrate to one.

This makes contact with Y1 Temperature and Matter, where the distribution in higher dimensions was discussed.